Objectives of OS

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- The user gives a command and the OS translates the command into a form that the machine can understand and execute.
- OS has two main objectives—
  - (1) to make the computer system convenient and easy to use, for the user, and
  - (2) to use the computer hardware in an efficient way, by handling the details of the operations of the hardware

Types of OS

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- OS are classified into different types depending on their capability of processing—

1) Single user,
2) Multiuser,
3) Multitasking,
4) Multiprocessing,
5) Real time, and
6) Embedded.

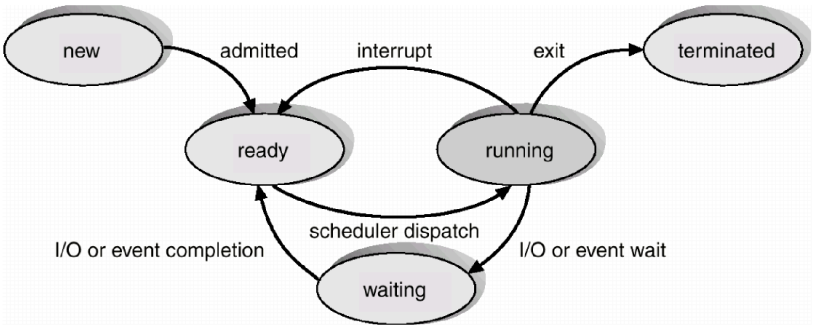
Function of OS

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- Process Management
- Memory Management
- File Management
- Device Management
- Protection and Security
- User Interface and Command Interpreter

Process State

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CPU Scheduling

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- Different types of CPU Scheduling Algorithms:
  - First Come First Serve (FCFS)
  - Shortest Job First (SJF)
  - Round Robin (RR)
- Pre-emptive:
  - CPU can withdraw any process from execution.

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Process Synchronization

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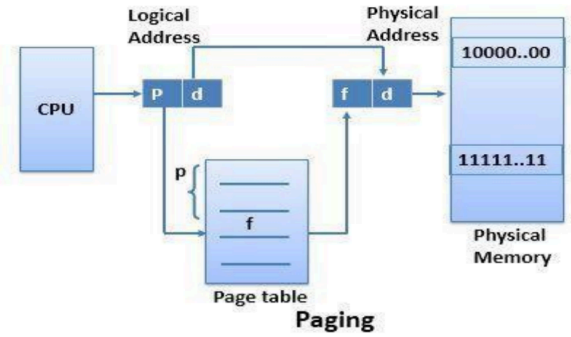
- Concurrent Processes
  - Two or more process independent of each other execute at the same time.
- Race Condition
  - Multiple process accessing and manipulating same data and the final result depends on the order of execution.

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Memory Management

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- Paging



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File Management

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- A set of files and
- A directory structure

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Device Management

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- OS manages and controls the devices attached to the computer.
- The OS communicates with the I/O hardware via the device driver software. The device driver software comes along with each device

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